

MSc(IT) Semester III [Autumn 2024-25]

IT644: Web Services & SOA

15-Oct-2024

Duration: 1 hour

Max. Marks: 40

Roll Number _____ Name _____

1. Summarize types of cloud services.

(2 Marks)

Ans:

2. What is technical debt? Summarize its classification.

(2 Marks)

Ans:

3. Narrate ways that we could use to implement secured authentication/authorization in microservices-based web applications? (2 Marks)

Ans:

4. Summarize core principles of REST, and briefly summarize significance of one of it. (2 Marks)

Ans:

5. Define: Rate Limiting. Summarize ways to handle it.

(2 Marks)

Ans:

6. Give comparative analysis of SOAP and REST based-on following parameters: protocol, performance, security, data format, and state scenario.

(2 marks)

Ans:

7. Summarize UDDI data model using a suitable diagram.

(2 Marks)

Ans:

8. Differentiate stateful and stateless architecture giving one example for each.

(2 Marks)

Ans:

9. Summarize core components of WSDL.

(2 Marks)

Ans:

10. Summarize tools and frameworks for SOAP based development as discussed in our classroom/lab. (2 marks)

Ans:

11. What type of hybrid architecture is used in today's web applications? Give one example. (2 Marks)

Ans:

12. Summarize key components of SOAP architecture. (2 marks)

Ans:

13. List key steps followed for monolith to microservices migration.

(2 marks)

Ans:

14. Summarize microservices key uses cases based-on its perceived benefits.

(2 Marks)

Ans:

15. Summarize benefits of GoLang, in particular performance improvement as in Bitly.com case-study. (2 marks)

Ans:

16. Which of the following are valid uses of Goroutines in Golang? Mark all that is applicable. (1 mark)
- (A) Running a function concurrently using the go keyword.
 - (B) Waiting for Goroutines to finish using sync.WaitGroup
 - (C) Sending data between Goroutines using mutexes.
 - (D) Managing synchronization using channels.
17. Which of the following are correct about Go slices? Mark all that is applicable. (1 mark)
- (A) Slices are reference types.
 - (B) Slices can be resized dynamically.
 - (C) Slices always have the same underlying array size.
 - (D) Slices have a fixed capacity.
18. Choose the correct explanation for the Golang code (see next page) behavior: (1 mark)
- (A) The ticker sends a signal every second, and the program prints the current time until the ticker is stopped after 5 seconds.
 - (B) The ticker sends a signal every millisecond, and the program prints the current time until it is stopped after 5 seconds.

(C) The ticker sends a signal every second, but the program doesn't print anything because the ticker is not running.

(D) The ticker sends a signal every second, but the program runs indefinitely without stopping.

```
package main

import (
    "fmt"
    "time"
)

func main() {
    // Create a new ticker that ticks every second
    ticker := time.NewTicker(1 * time.Second)
    done := make(chan bool)

    // Goroutine to listen to ticker events
    go func() {
        for {
            select {
            case <-done:
                return
            case t := <-ticker.C:
                fmt.Println("Tick_at", t)
            }
        }
    }()

    // Let the ticker run for 5 seconds
    time.Sleep(5 * time.Second)
    ticker.Stop() // Stop the ticker
    done <- true  // Signal the goroutine to stop
    fmt.Println("Ticker_stopped")
}
```

19. What is not correct from the following about a select statement in Golang?

(1 mark)

- (A) It allows waiting on multiple channel operations.
- (B) It can have a default case for non-blocking behavior.
- (C) It can be used with buffered and unbuffered channels.
- (D) It ensures fairness by always choosing the first ready case.

20. What is the output of following Golang code? Reason it to get credit for it. (1 mark)

```
package main

import "fmt"

type Data struct {
    value int
}

func main() {
    d := Data{10}
    updateValue(d)
    fmt.Println(d.value)
}

func updateValue(d Data) {
    d.value = 20
}
```

Ans:

21. Which of the following are valid ways to handle errors in Golang? Mark all that is applicable. (1 mark)
- (A) Returning an error value from a function.
 - (B) Using defer to handle errors after panic.
 - (C) Implementing a try-catch block for error handling.
 - (D) Wrapping errors using `fmt.Errorf()` or `errors.Wrap()`
22. What from following is not correct about defer statement in Golang? Mark all that is applicable. (1 mark)
- (A) Deferred functions are executed after return statements.
 - (B) Deferred functions can alter the return values of a function.
 - (C) Deferred functions are executed when a Goroutine is stopped.
 - (D) Deferred functions are executed in First In, Last Out (FILO) order.

23. What is the output of following Golang program using closures? Reason it to get credit for it. (1 mark)

```
package main

import "fmt"

func counter() func(int) int {
    sum := 0
    return func(x int) int {
        sum += x
        return sum
    }
}

func main() {
    count := counter()
    fmt.Println(count(10))
    fmt.Println(count(20))

    count1 := counter()
    fmt.Println(count1(30))
}
```

Ans:

24. What is the output of following Golang program? Reason it to get credit for it. (1 mark)

```
package main

import "fmt"

func main() {
    a := []int{1, 2, 3}
    b := append(a, 4, 5)
    fmt.Println(a, b)
}
```

Ans:

25. What is the output of following Golang program? Reason it to get credit for it. (1 mark)

```
package main

import "fmt"

func main() {
    var a []int
    fmt.Println(a == nil)
}
```

Ans: